Heritage Bank — Retail Banking Team: Technology Context

Services Owned by This Team

Service	Responsibility	Technology
Account Service	Account overview, balances (via BIH), transaction history, statement retrieval	Spring Boot 3.2 → BIH → AS/400
Transfer Service	Internal transfers, ACH external, Zelle, RTP, Bill Pay, wire initiation	Spring Boot 3.2 → BIH / EWS / TCH
Card Management Service	Debit card lock/unlock, travel notice, spending controls, dispute initiation (Reg E)	Spring Boot 3.2 → FIS Worldpay API
Notification Preferences Service	Customer alert settings; channel preferences (push/SMS/email)	Spring Boot 3.2 + PostgreSQL
Account Opening Service	Blend integration; identity verification flow; CIP document handling; BIH account creation	Spring Boot 3.2 + Blend API + BIH

Frontends Owned by This Team

Client	Technology	Notes
Mobile Banking App	React Native 0.74 (iOS 16+, Android 10+)	Okta OIDC/PKCE auth; biometric (FaceID/TouchID) as second factor; offline read of cached balance
Online Banking Portal	React 18 SPA (TypeScript)	Okta session-based auth; hosted on CloudFront + S3

Key Technical Constraints

AS/400 Batch Window Handling

- The Account Service and Transfer Service must **queue all AS/400 write operations** when the batch window is active (midnight–4 AM CT, 6 PM CT EOD)
- During batch windows: reads are allowed (AS/400 serves book balance, which is displayed with a "last updated" timestamp); writes are queued in IBM MQ and processed when the window closes
- Customer-facing UIs must show a "processing" status for queued operations without alarming the customer

Balance Freshness

- Available balance is fetched from the FiServ Open Solutions API on every page load (not cached for more than 30 seconds)
- ElastiCache Redis caches the available balance with a **30-second TTL** to limit AS/400 API call volume during high-traffic periods

- If the balance API returns a 5xx error, the UI shows the last cached balance with a "balance as of [timestamp]" label

Idempotency

- All ACH and RTP operations use an idempotency key (UUID v4 generated client-side, persisted in the Transfer Service before submission to BIH)
- Duplicate submissions (e.g., double-tap on the Submit button) return the result of the original operation without re-submitting to AS/400

Card Management

- FIS Worldpay API lock/unlock is synchronous (response within 3 seconds); Card Management Service reflects the new state in the mobile app within 3 seconds of the API response
- Transaction disputes are stored in the Card Management Service database; provisional credit is applied by the Fraud Analyst team (manual process in Phase 1); automated provisional credit in Phase 2

External Integrations (this team)

Integration	Protocol	Direction	Notes
BIH → AS/400	FiServ Open Solutions REST + IBM MQ	Account/Transfer Service → BIH → AS/400	All AS/400 operations via BIH only
FIS Worldpay	REST API	Card Management Service ↔ Worldpay	Lock/unlock, travel notice, spending controls; tokenized card data
EWS / Zelle	EWS REST API	Transfer Service → EWS	P2P payments; 15-min hold on first recipient
TCH RTP	TCH RTP REST API	Transfer Service → TCH	Real-time payment submission; no rollback once submitted
Fiserv CheckFree	REST API	Transfer Service → CheckFree	Bill pay disbursement; Heritage sends payee + amount, CheckFree disburses
Blend	Blend API	Account Opening Service → Blend	Consumer account opening; KBA; CIP document upload
Okta	OIDC + SAML	Mobile + Web ← Okta	Auth, MFA enrollment, session management
NICE Actimize	Kafka (consumer)	Actimize → Notification Service	Fraud alerts consumed from Kafka and pushed to customer

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Non-Functional Requirements (this team)

Category	Requirement

Balance refresh	Available balance cached max 30 seconds; always shows real-time label with "as of [time]"
Card lock/unlock	FIS Worldpay API response within 3 seconds; reflected in app within 3 seconds
Zelle P2P	First payment to new recipient: 15-minute hold enforced by Transfer Service (not EWS)
ACH cut-offs	Same-day ACH (SDIE): 2:30 PM CT; next-day ACH: 8 PM CT — Transfer Service enforces these, rejecting late submissions
Session security	10-minute inactivity timeout (web); 5-minute (mobile); auto-logout with clear session and Okta token revocation
Availability	99.9% for Account Service and Transfer Service; planned maintenance only in batch windows (midnight–4 AM CT)
Accessibility	WCAG 2.1 AA on mobile and web; VoiceOver / TalkBack tested in CI pipeline